The Divergence and Bhattacharyya Distance Measures in Signal Selection

THOMAS KAILATH, MEMBER, IEEE

REFERENCE: Kailath, T.: THE DIVERGENCE AND BHATTACHARYYA DISTANCE MEASURES IN SIGNAL SELECTION,[1] Stanford University, Stanford, Calif., and Stanford Research Institute, Menlo Park, Calif. Rec'd 4/6/66; revised 10/14/66. Paper 19TP67-906. IEEE TRANS. ON COMMUNICATION TECHNOLOGY, 15-1, February 1967, pp. 52–60.

ABSTRACT: Minimization of the error probability to determine optimum signals is often difficult to carry out. Consequently, several suboptimum performance measures that are easier than the error probability to evaluate and manipulate have been studied. In this partly tutorial paper, we compare the properties of an often used measure, the divergence, with a new measure that we have called the Bhattacharyya distance. This new distance measure is often easier to evaluate than the divergence. In the problems we have worked, it gives results that are at least as good as, and are often better, than those given by the divergence.

KEYWORDS: Communication Theory, Detection, Digital Signals, Fading, Matched Filters, Signal-to-Noise Ratio.

IN PROBLEMS of communication and radar, the optimum signals are those that minimize the probability of error. However, in many cases, direct minimization of the probability of error so as to determine an optimum signal set is often impossible. This may be because an explicit analytical expression for the error probability is too difficult to find, or even if it can be found, the expression may be too complicated for analytical or numerical minimization. Therefore, it is useful to search for signal selection criteria that may be weaker than the error probability but that are easier to evaluate and manipulate. Maximizing a *deflection ratio* or a *signal-to-noise* (*SNR*) is an example of such a weaker criterion—in general, maximizing the SNR may not lead to the minimum error probability, but the SNR may still be used because it is easier to analyze and does provide some guidance in many cases. In the search for suitable criteria, the notion of a *distance* between two probability distributions is quite useful. The two distributions will be the distributions of the observations in a binary hypothesis testing situation and the further *apart* we can make these distributions, hopefully the smaller will be the probability of mistaking one for the other. Therefore, various distance measures have been studied as simpler substitutes for the error probability.

What we would like for such distance measures is a property of the following type: if the distance between the two distributions is greater for a signal set α than for a signal set β, then the error probability for set α is always less than the error probability for set β. This is too much to hope for—we shall only be able to find *weaker* relations between distance measures and the probability of error.

In statistics, the use of such distance measures has a long history beginning with the work of Pearson (as quoted in Tildesley [1]). Two widely used distance measures in statistics are the D^2-statistic of Mahalanobis [2], [3] and the linear discriminant function introduced by Fisher [4]. These measures are discussed in detail in many statistical texts, e.g., Anderson [5] or Rao [6]. With the advent in 1948 of Shannon's information theory, the divergence, a measure closely related to Shannon's logarithmic measure of information, has become popular, though it had first been proposed before Shannon's work by Jeffreys [7]. Several properties and applications of the divergence are noted in Kullback [8].

In the engineering literature the divergence has recently been applied to several problems with varying degrees of success (as seen in Marill and Green [9], Grettenberg [10], Hingorani [11], and others). In this paper, we propose another criterion for signal selection based on a statistic first introduced in a statistical context[2] by Bhattacharyya [12]. In the various signal selection problems we have considered so far, the Bhattacharyya distance has turned out to give better results than the divergence. Therefore, it seems like a useful measure to study.

It should be pointed out that the Bhattacharyya distance is actually a special case of a more general distance measure introduced by Chernoff [15]. This measure, which will be discussed briefly in Section III, is in general *closer* to the error probability than the Bhattacharyya distance. On the other hand, it is usually not as easy to evaluate as the Bhattacharyya distance, which, for this reason, will be preferred in this paper.

We shall first briefly review some of the properties of the divergence and of the Bhattacharyya distance and shall then compare the two measures in various ways.

We shall confine ourselves largely to two-hypothesis problems. The two hypotheses will be denoted h_i, $i = 1,2$; and we shall set $p_i(x)dx$ = probability of the observation x when hypotheses h_i is true. The observation x will, for simplicity be taken as an N-component vector. [The concepts can be extended to the case of continuous-time observations (See Section IV).]

The likelihood ratio

$$L(x) = \frac{p_1(x)}{p_2(x)}$$

[1] This work was supported by AFOSR Contract AF 49(638)1517, by Tri-Service Nonr 225(83), and by DA 36-039SC-90859 at S.R.I.

[2] Another early and well-known statistical application of this statistic was made by Kakutani [13], who notes an earlier appearance in a nonstatistical problem (Hellinger [14]). Therefore, the names Hellinger and Kakutani are often associated with the Bhattacharyya statistic.

Authorized licensed use limited to: UNIVERSIDAD CARLOS III MADRID. Downloaded on January 30,2026 at 11:50:40 UTC from IEEE Xplore. Restrictions apply.

will play a prominent part in our discussions. In fact, both the divergence and the Bhattacharyya distance are convex functionals of the likelihood ratio.

I. The Divergence and the Bhattacharyya Distance

The divergence was first introduced by Jeffreys [7], [16]. It is defined as the difference in the mean values of the log-likelihood ratio under the two hypotheses

$$J = E_1[\ln L(x)] - E_2[\ln L(x)] \tag{1}$$

where

$$E_i[\ln L(x)] = \int [\ln L(x)] p_i(x) dx, \; i = 1,2. \tag{2}$$

These expectations of $\ln L(x)$ are often called the Kullback-Leibler (K-L) numbers (Kullback and Leibler [17] and written

$$I(1, 2) = E_1[\ln L(x)], \; I(2, 1) = -E_2[\ln L(x)]. \tag{3}$$

In general

$$I(1, 2) \neq I(2, 1). \tag{4}$$

The divergence is a symmetrized form of the K-L numbers

$$J = I(1, 2) + I(2, 1). \tag{5}$$

We shall calculate J for several examples in Section II. The divergence satisfies all the postulates for a metric (distance) except the triangle inequality. That is we have $J(1, 2) = J(2, 1)$, $J(1, 2) \geq 0$ with equality only when $p_1 = p_2$ but $J(1, 2) + J(2, 3)$ may be less than $J(1, 3)$. This last statement can be checked by taking p_1, p_2, p_3 to be normal distributions with means zero and variances 1, 4, and 5.

Several other properties of the divergence—its additivity for independent observations, its behavior under transformations, relations to channel capacity, etc.,—and various applications to classification and hypothesis testing are given in Kullback [8].

For the present paper, we are interested in a property of the divergence that is related to signal selection. If we have two signal sets—or more generally, two sets of parameters for the densities $p_i(x)$, $i = 1,2$—say α and β, we can *order* (or rank) them by means of the divergence. That is, we can say the signal set α is better than the signal set β if $J(\alpha)$, the divergence between p_1 and p_2 for parameters α is larger than $J(\beta)$. In communications and radar, the phrase "signal set α is better than set β" generally means "the error probability (or more generally, the Bayes' risk) is less for signal set α than it is for signal set β." Now the error probability depends on the total distribution (and hence all the moments) of the likelihood ratio while the divergence depends only upon the first moments (mean values) of the likelihood ratio. Therefore, in general, if $J(\alpha) > J(\beta)$ [i.e., α is better than β under the divergence criterion] it will not be true that $P_e(\alpha) < P_e(\beta)$. Consequently, the following theorem of Karlin and Bradt [18] is quite surprising.

If $J(\alpha) > J(\beta)$, there exists a set of prior probabilities $\pi = (\pi_1, \pi_2)$ for the two hypotheses, for which

$$P_e(\alpha, \pi) < P_e(\beta, \pi).^3 \tag{6}$$

This is an interesting result, but it would be much more so if we could assert something more than just existence about the *good* set of prior probabilities. Unfortunately, this is hard to do. Nevertheless, Karlin and Bradt's result encouraged Grettenberg [10] to apply the divergence criterion to several signal selection problems. He was followed by several others.

Before discussing these signal selection problems, let us introduce the Bhattacharyya distance. We first define the Bhattacharyya coefficient for two densities $p_i(x)$ by

$$\rho = \text{Bhattacharyya coefficient} = \int \sqrt{p_1(x) p_2(x)}\, dx. \tag{7}$$

Clearly ρ lies between zero and unity. We can associate several distance measures with this coefficient. The one we shall use will be[4]

$$B = \text{the Bhattacharyya distance} = -\ln \rho. \tag{8}$$

Clearly $0 \leq B \leq \infty$. It can be shown that B need not obey the triangle inequality (consider normal populations with means zero and variances 1, 4, and 5). (But $\sqrt{1 - \rho}$ does obey the triangle inequality.) We shall evaluate B for several distributions in Sections II and III.

The Bhattacharyya distance has many interesting properties, some of which we shall discuss later, but what led us to investigate its application to signal selection was the fact that B has the property discovered by Karlin and Bradt for the divergence.

If for two sets of system parameters α and β, we have $B(\alpha) > B(\beta)$ or equivalently $\rho(\alpha) < \rho(\beta)$, then there exists a set

$$\pi = (\pi_1, \pi_2) \tag{9}$$

of prior probabilities for which $P_e(\alpha, \pi) < P_e(\beta, \pi)$.

This result, as well as the earlier one of Karlin and Bradt for the divergence, follows easily from a theorem proved by Blackwell [19] and attributed by him to unpublished work by H. Bohnenblust, L. Shapley, and S. Sherman. We have

Theorem (Blackwell)[5]

$P_e(\alpha, \pi)$ will be less than or equal to $P_e(\beta, \pi)$ for all π if and only if

$$E_\alpha[\psi(L_\alpha) | h^{(2)}] \leq E_\beta[\psi(L_\beta) | h^{(2)}] \tag{10}$$

for all continuous concave functions $\psi(L)$. $\{E_\alpha[\psi(L_\alpha)|h^{(2)}]$ is the expectation of $\psi(L_\alpha)$ under the hypothesis $h^{(2)}.\}$

[3] $P_e(\alpha, \pi)$ is the error probability with parameter set α and prior probabilities Π.

[4] Another possibility is the quantity $\sqrt{1-\rho}$.

[5] We note incidentally that this theorem is attributed to Karlin and Bradt by Grettenberg (cf. Theorem 1 of Grettenberg [10]), whereas Grettenberg's Theorem 2 should in fact have been credited to Karlin and Bradt.

Authorized licensed use limited to: UNIVERSIDAD CARLOS III MADRID. Downloaded on January 30,2026 at 11:50:40 UTC from IEEE Xplore. Restrictions apply.

The proof of (9) is now simple. We note that $\sqrt{L}$ is a concave function of L and that

$$\rho(\alpha) = \int \sqrt{p_1(x, \alpha)p_2(x,\alpha)}dx$$

$$= \int \sqrt{\frac{p_1(x, \alpha)}{p_2(x, \alpha)}}\, p_2(x, \alpha)dx = E_\alpha[\sqrt{L_\alpha}|h^{(2)}]. \quad (11)$$

Similarly

$$\rho(\beta) = E_\beta[\sqrt{L_\beta}|h^{(2)}].$$

But then $\rho(\beta) < \rho(\alpha)$ implies $E_\beta[\sqrt{L_\beta}|h^{(2)}]\, E_\alpha[\sqrt{L_\alpha}|h^{(2)}]$, which contradicts (10). Therefore, Blackwell's theorem shows that if $\rho(\beta) < \rho(\alpha)$, then $P_e(\alpha, \pi)$ cannot be strictly smaller than $P_e(\beta, \pi)$ for all π, which is just the result (9).

II. Some Applications to Signal Selection

Grettenberg [10] used the comparison property (6) of the divergence J as a rationale for using J as a tool for signal selection. We have found that the Bhattacharyya distance B has a similar property (9) and this leads us to compare J and B as tools for signal selection. We shall consider three examples originally studied by the divergence method—two were treated by Grettenberg [10] and one by Reiffen and Sherman [20].

A. Gaussian Processes with Unequal Mean Value Functions

The detection problem is to choose between the two hypotheses

$$h_1: x(t) = m_1(t) + n(t),\ h_2: x(t) = m_2(t) + n(t),\ t \epsilon T$$

where $m_1(t)$ and $m_2(t)$ are known functions of time, $n(t)$ is a sample function of a zero mean Gaussian process with covariance function $R(t, s)$, and $x(t)$ is the observation on the basis of which we are to decide whether h_1 or h_2 is true. For simplicity, we shall assume that the noise is white, with

$$R(t, s) = \frac{N_0}{2}\delta(t - s)$$

and that the signals $m_i(t)$, $i = 1,2$ have equal energies

$$\int_T m_i^2(t)dt = E,\ i = 1,2 \quad (12)$$

and a correlation coefficient μ defined by

$$\int_T m_1(t)m_2(t)dt = E\mu. \quad (13)$$

Now some computation will show that

$$B = \frac{1}{8}J = \frac{1}{8}\frac{2E}{N_0}(1 - \mu) \quad (14)$$

so that both the Bhattacharyya distance B and the divergence J are maximized by choosing the signals $m_i(t)$, $i = 1,2$ to be antipodal, viz., such that

$$m_1(t) = -m_2(t) \text{ or } \mu = -1. \quad (15)$$

It is well-known that this choice of signals is in fact the one that minimizes the error probability for all prior probabilities II, so that in this problem both the divergence and the Bhattacharyya distance signal selection criteria yield signals that are in fact also optimum on an error probability basis. This happy coincidence is unfortunately not universal, as the next example shows.

B. Gaussian Processes with Unequal Covariance Functions

The problem is the following: one of two signals $m_i(t)$, $i = 1,2$ each of energy E/L

$$\int_T m_i^2(t)dt = E/L,\ i = 1,2 \quad (16)$$

is transmitted over L channels in each of which the signal is perturbed multiplicatively by α_k and additively by $n_k(t)$, so that the received signals are

$$h^{(i)}: x_k(t) = \alpha_k m_2(t) + n_k(t),\ k = 1,\ldots,L,\ i = 1,2.^6 \quad (17)$$

We shall assume that the $\{\alpha_k\}$ are independent identically distributed Gaussian random variables with

$$E[\alpha_k] = 0,\ E[\alpha_k\alpha_j] = \sigma^2\delta_{kj},\ k, j = 1,\ldots,L. \quad (18)$$

The additive noises will be assumed to be Gaussian, independent, identically distributed with

$$E[n_k(t)] = 0,\ E[n_k(t)n_j(s)] = \left[\frac{N_o}{2}\,\delta(t - s)\right]\delta_{kj} \quad (19)$$

and independent of the random variables $[\alpha_k]$

$$E[\alpha_k n_j(t)] = 0,\ k, j = 1,\ldots,L. \quad (20)$$

The signal selection problem in the above is the following: if we had just a single link, there is a large probability that the transmitted signal will be drastically weakened by the multiplicative noise assuming a low value. To combat this loss of signal, which is usually called the loss due to *fading*, a classical solution (1923–1927) is to use a number of links to provide *diversity*. If we use a number of links with independent multiplicative noises, the probability that the signal will be simultaneously reduced on all links can be made quite small. It would therefore seem desirable to use as many diversity links as possible. However, another factor has to be taken into account. The total signal energy is usually constrained, say to E, and therefore, the more links we use the less will be the available signal energy per link. If too many links are used, the signal energy per link may be so small that the resulting greater degradation in the performance of each link cannot be compensated for by the diversity gain provided by the use of the larger number of links. There should therefore be an optimum number of links to use for a given total signal energy E.

Let us now see what results the use of the divergence and Bhattacharyya measures yield. First for the diver-

[6] In the communication problem, it is physically most reasonable to take α_k, $x_k(t)$, $m_2(t)$ and $n_k(t)$ to be the complex envelopes of the corresponding real (narrowband) signals. For comparison with the earlier results of Pierce [21] and Grettenberg [10], we shall make this assumption here. However B and J can be calculated and optimized whether or not the signals are narrowband.

Authorized licensed use limited to: UNIVERSIDAD CARLOS III MADRID. Downloaded on January 30,2026 at 11:50:40 UTC from IEEE Xplore. Restrictions apply.

gence, some computation shows that, if (for simplicity) we make the additional assumption that the signals are orthogonal, viz.,

$$\int_0^T m_1(t)m_2(t)dt = 0 \tag{21}$$

then the divergence is

$$J = \frac{R^2}{R + L}, R = 2E\sigma^2/N_o. \tag{22}$$

And clearly for a fixed E, this is maximized by setting $L = 1$, that is by putting all the energy in a single link. However, our previous discussion indicates that this is not a very reasonable thing to do when R is large. And in fact, by the use of certain bounding techniques and by numerical evaluation, Pierce [21] found the optimum number of links as a function of the SNR, $R = 2E\sigma^2/N_o$, for the case of equal prior probabilities. Pierce showed that for low SNR $R \ll 1$ the use of a single link is optimum but that for large SNR, $R \gg 1$, the optimum number of links is given (to a very good approximation) by the remarkably simple formula[7]

$$L_{\text{opt}} = [R/3] \tag{23}$$

where $[x]$ means "the greatest integer less than or equal to x." We conclude from this that in Grettenberg's words ([10], p. 275), "the range of source statistics for which the maximum divergence code has a lower error probability than a different code does not always include the equiprobable case." We should repeat that the Karlin-Bradt theorem (6) tells us that the maximum divergence solution ($L = 1$) must be best, no matter how large the SNR is, for some set of prior probabilities. Unfortunately, the theorem gives us no information about this *good* set of prior probabilities and in many problems it might be that this set is highly skewed [π_1 = unity] and therefore highly improbable.

The Bhattacharyya distance in this problem can be calculated to be

$$\rho = e^{-B} = \left(1 + \frac{R}{L}\right)^L \Big/ \left(1 + \frac{R}{2L}\right)^{2L}, R = \frac{2E\sigma^2}{N_o}.$$

Differentiation shows that B is maximum when

$$x^2/(x + 1)(x + 2) = \ln\{(x^2 + 4x + 4)/4x + 4\}, x = R/L$$

and computer solution of this equation yields

$$L_{\text{opt}} = R/3.07$$

which is the same as the solution (23) found by Pierce [21] from the exact expression for P_e, which incidentally is

$$P_e = \frac{1}{(2 + R/L)^L} \sum_{k=0}^{L-1} \binom{L - 1 + k}{k} \left(\frac{1 + R/L}{2 + R/L}\right)^k.$$

[7] Grettenberg ([10], p. 275) claims erroneously that the error probability for this case decreases monotonically with L, so that the best choice is $L = \infty$.

It can be shown [see discussion after (49)] that asymptotically the Bhattacharyya distance essentially determines the error probability in the sense that

$$\lim_{P_e \to 0} P_e = c \cdot e^{-B}, c = \text{a constant.}$$

Thus for the (zero-mean) Gaussian signal in Gaussian noise problem, the Bhattacharyya distance yields better results (at least for the situation of most interest in communication problems where $\pi_1 \doteq \pi_2$) than the divergence.

Gaussian processes with different means and covariance yield similar results—the divergence and Bhattacharyya distance criteria yield similar results at low SNR,[8] but at high SNR they give different solutions, with the divergence solution being poorer (at least for $\pi_1 \doteq \pi_2$).

However, as a last example, we treat a problem in which the two criteria again yield similar results.

C. Poisson Processes with Unequal Mean Value Functions

The detection problem is to choose between two Poisson processes whose mean value functions are

$$m^{(k)}(t) = \lambda_i^{(k)} + \lambda_o, (i - 1)\Delta \leq t \leq i\Delta, \quad i = 1, \ldots N, k = 1,2 \tag{24}$$

where the observation interval T has been divided into N intervals of length Δ, $N\Delta = T$. The $\{\lambda_i^{(k)}\}$ are determined by our choice of signals. We constrain this choice by

$$\lambda_i^{(k)} \geq 0, \Sigma\Delta(\lambda_i^{(1)} + \lambda_i^{(2)}) = 2E. \tag{25}$$

The observation is the sequence $\{n_i\}$ of the number of events (photon arrivals at a photon detector) occurring in the several intervals; the number of events in the ith interval is Poisson distributed with

$$P^{(k)}\{n_i \text{ events in } (i - 1)\Delta \leq t \leq i\Delta\} = \frac{[(\lambda_i^{(k)} + \lambda_o)\Delta]^{n_i}}{n_i!} e^{-\Delta}(\lambda_i^{(k)} + \lambda_o).^9 \tag{26}$$

The numbers of events in disjoint intervals are assumed to be independently distributed. For this problem, the divergence J, and Bhattacharyya distance B can be shown to be

$$J = \sum_1^N (\lambda_i^{(1)} + \lambda_o) \ln\left[\frac{\lambda_i^{(1)} + \lambda_o}{\lambda_i^{(2)} + \lambda_o}\right] + \sum_1^N (\lambda_i^{(2)} + \lambda_o) \ln\left[\frac{\lambda_i^{(2)} + \lambda_o}{\lambda_i^{(1)} + \lambda_o}\right]. \tag{27}$$

$$B = 2\sum_1^N [\sqrt{\lambda_i^{(1)} + \lambda_o} - \sqrt{\lambda_i^{(2)} + \lambda_o}]^2. \tag{28}$$

Let us first determine the choice of signals (i.e., of the $\lambda_i^{(k)}$) that maximizes J. A first step is clearly to make

$$\lambda_i^{(k)} = 0 \text{ when } \lambda_i^{(j)} \neq 0, j \neq k \tag{29}$$

[8] It is easy to check in this problem that at low SNR, $J \doteq 8B$. [At all SNR, we shall show, cf. (72), that for Gaussian densities $J \geq 8B$.]

[9] For convenience, we shall henceforth take $\Delta = 1$.

Authorized licensed use limited to: UNIVERSIDAD CARLOS III MADRID. Downloaded on January 30,2026 at 11:50:40 UTC from IEEE Xplore. Restrictions apply.

i.e., to make the signals *orthogonal* in a certain sense. However, the divergence can be increased still further. Thus, let us define a sequence

$$q_i = \lambda_i^{(1)} \text{ if } \lambda_i^{(2)} = 0,\ \lambda_i^{(1)} \geq 0. \tag{30}$$

The constraint (25) now is $\sum_1^N q_i = 2E$ and the divergence is

$$\frac{1}{\lambda_o} J = \sum \frac{q_i}{\lambda_o} \ln\left(\frac{q_i}{\lambda_o} + 1\right),\ \sum q_i = 2E. \tag{31, 32}$$

A stationary point for the expression in (32) is easily found by the use of Lagrange multipliers to be

$$q_i = 2E/N,\ i = 1,\ldots,N.$$

However, this stationary point is a minimum rather than a maximum. We can show that J is a convex function of the q_i so that the maximum of J will occur at any extreme point.

$$q_i = 2E \text{ for some } i,\ q_i = 0,\ i \neq j. \tag{33}$$

Therefore, an optimum choice of signals (based on the divergence criterion) is

$$\{\lambda_i^{(1)}\} = \{2E, 0,\ldots,0\},\ \{\lambda_i^{(2)}\} = \{0, 0,..,0\} \tag{34}$$

or any permutation of (44).

As we shall now show, maximization of B yields the same result. With formula (28) for B in mind, we note that

$$\left[\sqrt{\lambda_i^{(1)} + \lambda_o} - \sqrt{\lambda_i^{(2)} + \lambda_o}\right]^2 = (\lambda^{(1)} + \lambda_o)\left[1 - \sqrt{\frac{\lambda_i^{(1)} + \lambda_o}{\lambda_i^{(2)} + \lambda_o}}\right]^2 \leq (\lambda_i^{(1)} + \lambda_o)\left[1 - \sqrt{\frac{\lambda_o}{\lambda_i^{(1)} + \lambda_o}}\right]^2 \tag{35}$$

with equality if and only if $\lambda_i^{(2)} = 0$. The argument can be repeated with $\lambda^{(1)}$ in place of $\lambda^{(2)}$. Therefore, for a maximum of B, we must set

$$\lambda_i^{(k)} = 0 \text{ if } \lambda_i^{(j)} > 0,\ j \neq k. \tag{36}$$

This is also one of the conditions necessary for the maximization of the divergence J. Therefore, as in (30), let us define a sequence

$$q_i = \begin{cases} \lambda_i^{(1)} \text{ if } \lambda_i^{(2)} = 0,\ \lambda_i^{(1)} \geq 0 \\ \lambda_i^{(2)} \text{ if } \lambda_i^{(1)} = 0,\ \lambda_i^{(2)} \geq 0 \end{cases} \tag{37}$$

with this the Bhattacharyya distance (28) can be written

$$B = 2\sum_1^N [\sqrt{q_i + \lambda_o} - \sqrt{\lambda_o}]^2,\ \sum_1^N q_i = 2E \tag{38}$$

and we have

$$B \leq 2\left[\sum_1^N \sqrt{q_i + \lambda_o} - \sqrt{\lambda_o}\right]^2 \tag{39}$$

with equality if and only if all the terms in the sum (38) save one, say $\sqrt{q_i + \lambda_o} - \sqrt{\lambda_o}$, are equal to zero. Therefore an optimum choice of signals (based on the Bhattacharyya distance criterion) is

$$\{\lambda_i^{(1)}\} = \{2E, 0,\ldots,0\},\ \{\lambda_i^{(2)}\} = \{0, 0,\ldots,0\} \tag{40}$$

or any permutation thereof.

Thus the optimum signals under the divergence and Bhattacharyya distance criteria are the same. Some computer studies by Blumenthal (Stanford University) indicate that sharp pulse-like signals of the form (34) or (40) also minimize the error probability (for equally likely signals). We should point out that Reiffen and Sherman [20] suggested the optimality of signals like (34) or (40) at low SNR[10] by using essentially the divergence criterion. Abend [22] proved that similar signals maximized a suitably defined SNR at all signal levels.

III. Some Further Properties of B and J

In the last section, we have shown in several problems that the Bhattacharyya distance is a good measure for signal selection. In this section, we shall present some additional properties of B, and pursue the comparison with the divergence J a little further. This section is largely a compilation of several results culled from the statistical literature. We begin with a convenient geometric interpretation of B or, actually, $\rho = e^{-B}$.

Geometric Interpretation

Bhattacharyya [12] proposed that the numbers $\{\sqrt{p_i(x)}$, all allowable x, $i = 1,2\}$, be regarded as the direction-cosines of two vectors in the space of x. Alternatively, we can regard them as defining two points on the unit sphere (since $\int p_i(x)dx = 1$, $i = 1,2$) of x. The Bhattacharyya coefficient ρ can be regarded as the cosine of the angle between these two vectors, i.e., $\rho = \cos \Delta$, Δ = angle between the lines with direction cosines

$$\{\sqrt{p_i(x)}\}. \tag{41}$$

The angle Δ must clearly be between 0 and $\pi/2$ and therefore $0 \leq \rho \leq 1$. It is now natural to also consider the distance, say ∂, between the two points $\{\sqrt{p_i(x)}, i = 1,2\}$ on the unit sphere

$$\partial \triangleq \int [\sqrt{p_i(x)} - \sqrt{p_2(x)}]^2 dx. \tag{42}$$

Clearly

$$\partial = 4 \sin^2 \frac{\epsilon}{2} = 2(1 - \cos \Delta) = 2(1 - \rho). \tag{43}$$

The distance ∂ occurs in the work of Jeffreys [7], [16].

The Kolmogorov Variational Distance

Another distance measure closely related to ρ is the *variational* distance of Kolmogorov (see Adhikari and Joshi [23], p. 71); this is defined as

$$K(\pi) = \frac{1}{2}\int |\pi_1 p_1(x) - \pi_2 p_2(x)| dx \tag{44}$$

[10] At low SNR all proposed measures of distance seem to coincide.

Authorized licensed use limited to: UNIVERSIDAD CARLOS III MADRID. Downloaded on January 30,2026 at 11:50:40 UTC from IEEE Xplore. Restrictions apply.

where π_1 and π_2 are the prior probabilities of the hypotheses h_1 and h_2. Kraft [24] has obtained certain important inequalities between ρ and $K(\pi)$. His inequalities (extended to the case $\pi_1 \neq \pi_2$) are

$$\sqrt{1 - 4\pi_1\pi_2\rho^2} \geq 2K(\pi) \geq [1 - 2\sqrt{\pi_1\pi_2}\rho]. \quad (45)$$

The first inequality follows via Schwarz's inequality from

$$[\int |\pi_1 p_1 - \pi_2 p_2| dx]^2 \leq \int |\sqrt{\pi_1 p_1} - \sqrt{\pi_2 p_2}|^2 dx \cdot \times$$
$$|\sqrt{\pi_1 p_1} + \sqrt{\pi_2 p_2}|^2 dx = [1 - 2\sqrt{\pi_1\pi_2}p][1 + 2\sqrt{\pi_1\pi_2}\rho].$$

The second inequality follows from

$$\int |\pi_1 p_1 - \pi_2 p_2| dx \geq \int |\sqrt{\pi_1 p_1} - \sqrt{\pi_2 p_2}|^2 dx$$
$$= 1 - 2\sqrt{\pi_1\pi_2}\rho.$$

There is another useful way of writing $K(\pi)$. A standard argument in analysis shows that

$$K(\pi) = \max_A |\pi_1 Pr\{x \epsilon A | h_1\} - \pi_2 Pr\{x \epsilon A | h_1\}| \quad (46)$$

where the maximum is taken over all sets A in the space of x. This form for $K(\pi)$ shows that it is closely related to the (optimum) probability of error. For, if we define $A_i = \{x | h_i \text{ is chosen when } x \text{ is received}\}$

$$\begin{aligned} P_e &= \min_{A_1} \left\{ \pi_1 \int_{A_2} p_1(x)dx + \pi_2 \int_{A_1} p_2(x)dx \right\} \\ &= \pi_1 - \max_{A_1} \left\{ \left| \int_{A_1} \pi_1 p_1(x) - \pi_2 p_2(x)dx \right| \right\} \\ &= \pi_1 - \max_{A_1} |Pr\{x \epsilon A_1 | h_1\} - \pi_2 Pr\{x \epsilon A_1 | h_2\}| \\ &= \pi_1 - K(\pi). \end{aligned} \quad (47)$$

Upper and Lower Bounds on Error Probability

Combining (47) and the relations (45), we note that

$$2\pi_1 - \sqrt{1 - 4\pi_1\pi_2\rho^2} \leq 2P_e = 2\pi_1 - 2K(\pi) \leq 2\pi_1 - 1 + 2\sqrt{\pi_1\pi_2}\rho. \quad (48)$$

If $\pi_1 = \pi_2 = 1/2$, this becomes

$$\frac{1}{4}\rho^2 \leq \frac{1}{2}(1 - \sqrt{1 - \rho^2}) \leq P_e \leq \frac{1}{2}\cdot\rho.^{11} \quad (49)$$

The bounds (49) are not entirely satisfactory. As $\rho \to 0$ (low P_e) the difference between the upper and lower bounds tends to zero but their ratio tends to infinity. However, in an important special case the upper bound is *exponentially* best, i.e.,

$$\ln P_e \doteq -\ln \rho = B \text{ for } P_e \to 0.$$

This happens under the following circumstances (Chernoff [15]):

1) The components X_i of the observation $X' = X_1, \ldots, X_N$ are independent and identically distributed
2) $\int [p_1(x)]^{1-t}[p_2(x)]^t dx$ is symmetric in t
3) N is large.

We shall not prove this result here. We should note that similar results are encountered in the study of error probability bounds for discrete-memoryless-channels (Gallager [25]). The upper bound in (49) appears to be good even in some cases where the conditions just given are not met (see e.g., Example B in the last section), but no general statement seems to be possible.

The Divergence and Error Probability

The upper bound on P_e in terms of ρ is quite useful. No similar bound in terms of the divergence J appears to be generally true. However by using an inequality between ρ and J we can obtain a crude lower bound on P_e in terms of J. The basic inequalities (apparently first derived by Hoeffding and Wolfowitz [26]) are

$$2B = -\ln \rho \leq I(1, 2) \text{ and } 2B \leq I(2,1) \quad (50)$$

where $I(1, 2)$ and $I(2, 1)$ are the Kullback-Leibler(K-L) numbers defined earlier [cf. (2)-(3)]. From (50), we get

$$4B \leq J \text{ or } \rho \geq \exp -(J/4). \quad (51)$$

The proofs of (50) follow via Jensen's equality

$$-\frac{1}{2}I(1, 2) = \int p_1(x) \ln \sqrt{\frac{p_2(x)}{p_1(x)}}\, dx = E_1\left[\ln \sqrt{\frac{p_2(x)}{p_1(x)}}\right] \leq \ln [E_1(\sqrt{p_2(X)/p_1(X)})] = \ln \rho.$$

A crude lower bound on P_e now follows from (49) and (51)

$$P_e \geq \rho^2/8 \geq \frac{1}{8}e^{-J/2}. \quad (52)$$

Of course, in particular cases, e.g., when $p_1(\cdot)$ and $p_2(\cdot)$ are Gaussian, we can obtain tighter relations between ρ and J than are provided by (51). We shall note some of these special relations when we give explicit formulas for B and J below. To close the present discussion of error bounds, we note that the divergence and the K-L numbers are quite useful in providing lower bounds on the conditional error probabilities. These follow from two formulas given by Kullback [8].

$$I(1, 2) \geq P_{e1} \ln \left(\frac{P_{e1}}{1 - P_{e2}}\right) + (1 - P_{e1}) \ln \left(\frac{1 - P_{e1}}{P_{e2}}\right) \quad (53)$$

and

$$I(2, 1) \geq P_{e2} \ln \left(\frac{P_{e2}}{1 - P_{e1}}\right) + (1 - P_{e2}) \ln \left(\frac{1 - P_{e2}}{P_{e1}}\right). \quad (54)$$

Note that for a fixed value of $P_{e1}\{P_{e2}\}$, (53), $\{(54)\}$ provides a lower bound on $P_{e2}\{P_{e1}\}$. The RHS in (53) and (54

[11] If we assume $\pi_1 = \pi_2$ when in fact they are not, we can only increase the probability of error. Hence, the upper bound in (49) is true for any (π_1, π_2).

Authorized licensed use limited to: UNIVERSIDAD CARLOS III MADRID. Downloaded on January 30,2026 at 11:50:40 UTC from IEEE Xplore. Restrictions apply.

have interesting interpretations—they are the values of $I(1, 2)$ and $I(2, 1)$ for binomial distributions with parameters P_{e1} and $(1 - P_{e2})$. Tables for the calculations in (53) and (54) are given by Kullback [8], (see pp. 74–75 and pp. 378–379).

As with the Bhattacharyya distance, the bounds provided by (53)-(54) are *exponentially optimum* under certain conditions. Thus if $x' = \underbrace{x_1, \ldots, x_N}$ where the x_i, $i = 1, \ldots N$ are independent and identically distributed, it has been shown that

$$\text{for any value of } P_e^{(1)}, \lim_{N\to\infty} -\frac{\ln P_e^{(2)}}{N} = I(1, 2)$$

where

$$I(1, 2) = \int p_1(x) \ln \left[\frac{p_1(x)}{p_2(x)}\right] dx.$$

A similar statement holds with 1 and 2 interchanged. These results were first given by Stein (unpublished); a proof appears in Kullback [8], pp. 74–77. (See also Chernoff [27]). Daly [28] has applied these results to some problems of radar signal design.

Relation of ρ to Fisher's Information

In problems of parameter estimation, we usually have not just two but a continuum of hypotheses. For such problems, a special information measure, introduced by Fisher in 1925, has been used in the statistical literature, see, e.g., Kullback [8], p. 26. It turns out that the Bhattacharyya distance and the divergence reduce to the Fisher information in estimation problems. To show this, we now let $p(x|\theta)$ = a probability density function depending upon a real-valued parameter θ, $0 \leq \theta \leq 1$, say. Then returning to our geometric picture, (41), we can calculate

$$\cos \delta s = \int \sqrt{p(x|\theta)p(x|\theta + \delta\theta)}\,dx \doteq 1 - \frac{(\delta\theta)^2}{8} \int_{-\infty}^{\infty} \frac{[\dot{p}(x|\theta)]^2}{p(x|\theta)}\,dx \quad (55)$$

where $\dot{p}(x|\theta) = (d/dx)\dot{p}(x|\theta)$ and δs = actual length between the points with direction cosiness

$$\sqrt{p(x|\theta)} \text{ and } \sqrt{p(x|\theta + \delta\theta)}. \quad (56)$$

In the limit as δs and $\delta\theta$ tend to zero we get

$$\left(\frac{ds}{d\theta}\right)^2 = \frac{1}{4}\int_{-\infty}^{\infty} \frac{[\dot{p}(x|\theta)]^2}{p(x|\theta)}\,dx = -\frac{1}{4}\int_{-\infty}^{\infty} p(x|\theta)\left[\frac{\partial^2}{\partial\theta^2} \ln p(x|\theta)\right] d\theta \triangleq \frac{1}{4} \text{ (the Fisher measure of information).} \quad (57)$$

This derivation is due to Bhattacharyya [12]. A similar calculation for the divergence[12] can be found in Kullback [8], p. 55. The somewhat unexpected appearance of the Fisher information in certain singular detection problems has recently been noted by Shepp [29].

Explicit Formulas for B and J

We can calculate B explicitly for a large class of distributions—the exponential family. Before giving the general formula we list a few important special cases.

Multinomial Distributions

$$\text{If } p_1(x) = \sum_{j=1}^{N} p_j^{(i)}\delta(x - j), \text{ then } B = -\ln\left[\sum_1^N \sqrt{p_j^{(1)}p_j^{(2)}}\right]. \quad (58)$$

Poisson Distributions

$$\text{If } p_i(x) = e^{-m_i}\cdot\frac{(m_i)^n}{n!}\delta(x - n), \text{ then } B = \frac{1}{2}(\sqrt{m_1} - \sqrt{m_2})^2. \quad (59)$$

Univariate Gaussian Distributions

$$\text{If } p_i(x) = N(m_1, \sigma_1), \text{ then } B = \frac{1}{4}\frac{(m_1 - m_2)^2}{\sigma_1^2 + \sigma_2^2} + \frac{1}{2}\ln\left\{\frac{\sigma_1^2 + \sigma_2^2}{2\sigma_1\sigma_2}\right\}. \quad (60)$$

Multivariate Gaussian Distributions

$$\text{If } p_i(x) = N(m_i, R_i), \text{ then } B = \frac{1}{8}(m_1 - m_2)'R^{-1}(m_1 - m_2) + \frac{1}{2}\ln\left(\frac{\det R}{\sqrt{\det R_1 \cdot \det R_2}}\right) \quad (61)$$

where

$$2R = R_1 + R_2. \quad (62)$$

Huzurbazar [30] has calculated B for exponential densities.

Exponential Densities

$$\text{Let } p_i(x) = d(x)b(h_i)\exp\left[\sum_{j=1}^{r} V_j(h_i)U_j(x)\right], \; i = 1,2. \quad (63)$$

The $V_j(h_i)$ are independent functions of h_i, or more precisely, of the V parameters of the density $p_i(\cdot)$. Since the $b(h_i)$ also depend upon h_i, we can express the $b(h_i)$ in terms of the $V_j(h_i)$ as

$$b(h_i) = g\{[V_j(h_i)]\}. \quad (64)$$

[Note: g depends on the h_i only through the $V_j(h_i)$, $j = 1, \ldots r$]. Then the Bhattacharyya distance is

$$B = -\ln\rho, \; \rho = \frac{\sqrt{b(h_j)b(h_i)}}{g\left(\left\{\frac{1}{2}V_j(h_1) + \frac{1}{2}V_j(h_2)\right\}\right)}. \quad (65)$$

[12] A variety of distance measure seem to have a similar property for small variations.

Authorized licensed use limited to: UNIVERSIDAD CARLOS III MADRID. Downloaded on January 30,2026 at 11:50:40 UTC from IEEE Xplore. Restrictions apply.

The Divergence

Huzurbazar [30] also calculates the divergence J for the exponential densities

$$J = \sum_{j=1}^{r} [V_j(h_2) - V_j(h_1)][E_2V_j(x) - E_1V_j(x)] \quad (66)$$

where

$$J = -\sum_{j=1}^{r}\sum_{k=1}^{r} \frac{\partial^2}{\partial V_j \partial V_k} \times \{V_j(h_2) - V_j(h_1)\}\{V_k(h_2) - V_k(h_1)\}. \quad (67)$$

We give one special case because of its importance.

If $p_i(x) = N(m_i, R_i)$, then

$$I(1, 2) = \frac{1}{2} \ln \frac{\det R_2}{\det R_1} + \frac{1}{2} tr\, R_1[R_2^{-1} - R_1^{-1}] + \frac{1}{2} tr\, R_2^{-1}[m_1 - m_2][m_1 - m_2]' \quad (68)$$

and

$$J(1, 2) = \frac{1}{2} tr[R_1 - R_2][R_2^{-1} - R_1^{-1}] + \frac{1}{2} tr[R_1^{-1} + R_2^{-1}][m_1 - m_2][m_1 - m_2]'. \quad (69)$$

If the covariances are equal, say, $R_1 = R_2 = R$, we have

$$2I(1, 2) = J = tr\, R^{-1}[m_1 - m_2][m_1 - m_2]'.^{13} \quad (70)$$

We note that the Bhattacharyya distance in this case is

$$B = \frac{J}{8}. \quad (71)$$

Whether or not $R_1 = R_2$, we can establish the inequality [which is better by a factor of two than the general inequality (51)]

$$J \geq 8B. \quad (72)$$

The simplest proof of (72) is obtained by considering the case $R_1 = \Lambda$, a diagonal matrix with entries λ_i, $i = 1, \ldots N$ and $R_2 = I$, the identity matrix. (The general case can be obtained from this by using a nonsingular matrix to simultaneously diagonalize R_1 and R_2.) Then from (67) and (75) we will have, after some algebra,

$$8B - J = 2\sum \ln(1 + \lambda_i)^2/4\lambda_i - \sum(\lambda_i - 1)^2/2\lambda_i.$$

But from the inequality $\ln (1 + \lambda) < \lambda$, we have

$$\ln (1 + \lambda)^2/4\lambda = \ln (1 + [(\lambda - 1)^2)/4\lambda]) \leq (\lambda - 1)^2/4\lambda \quad (73)$$

so that, using this for each λ_i, we get

$$8B - J \leq 0.$$

Differential Equations for B and J

Recently there has been interest (Schweppe [31], Schweppe and Athans [32]), in applying the techniques of control theory to signal design. To do this, it is convenient to obtain differential equations (in continuous-time) for the performance criterion. Schweppe has used B and J for these criteria and obtained differential equations for them when the signal and noise process are projections of Gaussian-Markov processes. The detailed formulas are too complicated to reproduce here, but they can be found in Schweppe [31]. However, we may note that for Gaussian processes with unequal covariances the differential equations for J are so much more complicated than the equations for B that Schweppe does not put down the equations for J. To quote him, "readers who subscribe to the theory that 'the best answer is the simplest answer' may decide that the Bhattacharyya distance is superior to the divergence."

[13] This is the famous Mahalanobis D^2-statistic.

IV. Further Applications and Extensions to Continuous-Time

The Bhattacharyya distance has found several applications in classical statistics (see, e.g., the papers by Matusita [33], Hannan [34], Rao [35]. A recent application is by Stein [36], p. 17. Stein's result is that the increase in the Bayes risk in a decision problem when an incorrect prior distribution, say π', is used on the hypotheses is bounded (under sufficient regularity conditions) by a constant times the Bhattacharyya distance between π' and the true prior distribution π. (This result also extends to prior distributions on unknown parameters.)

We have treated only the two-hypothesis case. In the M-hypothesis case, it appears that the average Bhattacharyya distance

$$\bar{B} = \sum_{i,j}^{M} \pi^{(i)}\pi^{(j)} \int_{-\infty}^{\infty} \sqrt{p_i(x)p_j(x)}\, dx$$

would be a suitable measure for signal selection. This average measure will have the same *minimax* property discussed by Grettenberg [10] for the average divergence. Furthermore it appears (cf. Gallager [25]), (87)) that, for low P_e, $\bar{B}$ essentially determines the error probability, P_e, in the M-hypothesis case.

We close with some comments on the continuous-time case. The usual method of solving continuous-time problems is to start with an approximate discrete-time problem and then obtain the final answer as a limit of the solution to the discrete-time problem. The same procedure works here with one modification. If we set $X(t_i) = x_i$, $i = 1, \ldots N$, then the Bhattacharyya distance when observations are restricted to $\{t_i, i = 1, \ldots N\}$ is

$$B_N = -\ln \int \sqrt{p_{1N}(x)p_{2N}(x)}dx \quad (74)$$

where

$p_{iN}(x)$ — probability density function of the N random variables $x(t_i)$, $i = 1, \ldots N$. (75)

As $N \to \infty$, $\lim_{N\to\infty} p_{iN}(x)$ is generally not defined. However, the ratios

$$\lim_{N\to\infty} q_{iN}(x) = \lim_{N\to\infty} \frac{p_{iN}(x)}{p_{1N}(x) + p_{2N}(x)} \quad (76)$$

will be well behaved (if the sets $\{t_i\}$ are monotone increasing) as $N \to \infty$. Therefore, in the continuous-time case we

Authorized licensed use limited to: UNIVERSIDAD CARLOS III MADRID. Downloaded on January 30,2026 at 11:50:40 UTC from IEEE Xplore. Restrictions apply.

should define B as

$$B = -\ln \rho, \text{ where } \rho = \lim_{N\to\infty} \int \sqrt{q_{1N}(x)q_{2N}(x)}dx. \quad (77)$$

Similar arguments can be used to define J in the continuous-time case. With this modification, all the relations and properties that we have obtained in Sections II and III will hold also for the case of continuous-time observations. The relations

$$\frac{1}{8}\rho^2 \leq P_e \leq \frac{1}{2}\rho \quad (78)$$

provide a simple criterion for singular detection, i.e., being able to discriminate between two processes with arbitrarily low error probability. We see that a sufficient condition, first noted by Kraft [24], for (extreme) singular detection is that $\rho = 0$. The Bhattacharyya distance can also be used to provide a simple proof of a result of Hajek [37] that the detection problem for two *Gaussian* processes for which the divergence $J = \infty$ is singular. The proof follows essentially from the fact that in the Gaussian case we can supplement the lower bound (51) on $J (J \geq 4B)$ by an upper bound on J of the form $J \leq (\text{constant}) \cdot B$. This and some other aspects of the interplay between B and J in questions of singular detection will be discussed in a separate note.

References

[1] Tildesley, "A first study of the Burmese skull," *Biometrika*, vol. 13, pp. 176–262, 1921.
[2] P. C. Mahalanobis, "Analysis of race mixture in Bengal," *J. Asiat. Soc. (India)*, vol. 23, pp. 301–310, 1925.
[3] ——"On the generalized distance in statistics," *Proc. Nat'l Inst. Sci. (India)*, vol. 12, pp. 49–55, 1936.
[4] R. A. Fisher, "The use of multiple measurements in taxonomic problems," *Ann. Eugencis*, vol. 7, pp. 179–188, 1936; *Contributions to Mathematical Statistics*. New York: Wiley, 1950.
[5] T. W. Anderson, *An Introduction to Multivariate Statistical Analysis*. New York: Wiley, 1958.
[6] C. R. Rao, *Advanced Statistical Methods in the Estimation of Statistical Parameters*. New York: Wiley, 1952.
[7] H. Jeffreys, "An invariant form for the prior probability in estimation problems," *Proc. Roy. Soc. A.*, vol. 186, pp. 453–461, 1946.
[8] S. Kullback, *Information Theory and Statistics*. New York: Wiley, 1959.
[9] T. Marill and D. M. Green, "On the effectiveness of receptors in recognition systems," *IEEE Trans. on Information Theory*, vol. IT-9, pp. 11–17, January 1963.
[10] T. L. Grettenberg, "Signal selection in communication and radar systems," *IEEE Trans. on Information Theory*, vol. IT-9, pp. 265–275, October 1963.
[11] G. P. Hingorani and J. C. Hancock, "A transmitted reference system for communication in random or unknown channels, *IEEE Trans. on Communication Technology*, vol. COM-13, pp. 293–301, September 1965.
[12] A. Bhattacharyya, "On a measure of divergence between two statistical populations defined by their probability distributions," *Bull. Calcutta Math. Soc.*, vol. 35, pp. 99–109, 1943.
[13] S. Kakutani, "On equivalence of infinite product measures," *Ann. Math. Stat.*, vol. 49, pp. 214–224, 1948.
[14] E. Hellinger, "Neue begrundung der theorie quadrátischer formen von unendlichvielen veränderlichen," *J. für die Reine und angew. Math.*, vol. 36, pp. 210–271, 1909.
[15] H. Chernoff, "A measure of asymptotic efficiency for tests of a hypothesis based on a sum of observations," *Ann. Math. Stat.*, vol. 23, pp. 493–507, 1952.
[16] H. Jeffreys, *Theory of Probability*. Oxford University Press, 1948.
[17] S. Kullback and R. A. Leibler, "On information and sufficiency," *Ann. Math. Stat.*, vol. 22, pp. 79–86, 1951.
[18] S. Karlin and R. N. Bradt, "On the design and comparison of dichotomous experiments," *Ann. Math. Stat.*, vol. 27, pp. 390–409, 1956.
[19] D. Blackwell, "Comparison of experiments," *Proc. Second Berkeley Symp. on Probability and Statistics*. Berkeley, Calif.: University of California Press, vol. 1, pp. 93–102, 1951.
[20] B. Reiffen and H. Sherman, "An optimum demodulator for Poisson processes: photon source detectors," *Proc. IEEE*, vol. 51, pp. 1316–1320, October 1963.
[21] J. N. Pierce, "Theoretical limitations of frequency and time diversity for fading binary transmissions, *IRE Trans. on Communications Systems (Corres.)*, vol. CS-9, pp. 186–187, June 1961.
[22] K. Abend, "Optimum photon detection," *IEEE Trans. on Information Theory*, vol. IT-12, pp. 64–65, January 1966.
[23] B. P. Adhikari and D. D. Joshi, "Distance discrimation et resume exhaustif," *Publs. Inst. Statist.*, vol. 5, pp. 57–74, 1956.
[24] C. H. Kraft, "Some conditions for consistency and uniform consistency of statistical procedures," University of California Publications in Statistics, 1955.
[25] R. G. Gallager, "A simple derivation of the coding theorem and some applications," *IEEE Trans. on Information Theory*, vol. IT-11, pp. 3–18, January 1965.
[26] W. Hoeffding and J. Wolfowitz, "Distinguishability of sets of distributions," *Ann. Math. Stat.*, vol. 29, pp. 700–718, 1958.
[27] H. Chernoff, "Large-sample theory: parametric case," *Ann. Math. Stat.*, vol. 27, pp. 1–22, 1956.
[28] R. F. Daly, "Signal design for efficient detection in randomly dispersive media," SRI Tech. Rept. on Project 186531-144, 1965.
[29] L. A. Shepp, "Distinguishing a sequence of random variables from a translate of itself," *Ann. Math. Stat.*, vol. 36, pp. 1107–1112, 1965.
[30] V. S. Huzurbazar, "Exact forms of some invariants for distributions admitting sufficient statistics," *Biometrika*, vol. 42, pp. 533–537, 1955.
[31] F. Schweppe, "On the distance between Gaussian processes: the state space approach," *Information and Control*, 1967.
[32] F. Schweppe and M. Athans, "On the design of optimal modulation schemes via control-theoretic concepts I: formulation," to be published.
[33] K. Matusita, "Decision rules, based on the distance for problems of fit, two samples and estimation," *Ann. Math. Stat.*, vol. 26, pp. 631–640, 1955.
[34] J. Hannan, "Consistency of maximum likelihood estimation of discrete distributions," in *Contributions to Probability and Statistics*, I. Olkin, Ed. Stanford, Calif.: Stanford University Press, 1960.
[35] C. R. Rao, "Asymptotic efficiency and limiting information," *Proc. Fourth Berkeley Symp. on Probability and Statistics*. Berkeley, Calif.: University of California Press, vol. 1, pp. 531–545, 1961.
[36] C. Stein "Approximation of improper prior measures by prior probability measures," Department of Statistics, Stanford University, Stanford, Calif., Tech. Rept. 12, 1964.
[37] J. Hajek, "On a property of normal distribution of any stochastic process" (in Russian), *Czech. Math. J.*, vol. 83, pp. 610–618, 1958; a translation appears in "Selected Translations" in *Math. Statistics and Probability*, vol. 1, pp. 245–252.

✤



Thomas Kailath (S'57–M'62) was born in Poona, India, on June 7, 1935. He received the B.E. degree from the College of Engineering, Poona, in 1956, and the M.S. and Ph.D. degrees in electrical engineering from Massachusetts Institute of Technology, Cambridge, in 1959 and 1961, respectively.

He worked at the Jet Propulsion Laboratories in Pasadena, Calif., until December, 1962. Since then he has been at Stanford University, Calif., where he is Associate Professor of Electrical Engineering. He is a consultant for several industrial companies and is also editor of the Prentice-Hall series in Information Theory.

He is a member of IMS, AMS, MAA, URSI, and Sigma Xi.

Authorized licensed use limited to: UNIVERSIDAD CARLOS III MADRID. Downloaded on January 30,2026 at 11:50:40 UTC from IEEE Xplore. Restrictions apply.